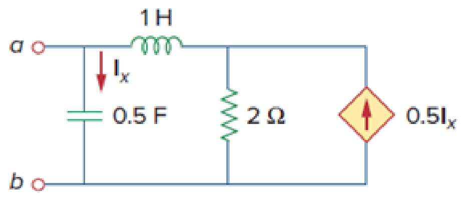
Chapter 14, Problem 80P

Problem

(a) For the circuit in Fig. 14.97, draw the new circuit after it has been scaled by *Km* = 200 and *Kf* = 104.

(b) Obtain the Thevenin equivalent impedance at terminals *a*-*b* of the scaled circuit at *ω* = 104 rad/s.

Figure 14.97:



Step-by-step solution

Step 1 of 8

(a)

Refer to Figure 14.97 in the textbook.

Consider that the circuit is scaled by the following scaling factors:

$$K_m = 200$$
$$K_f = 10^4$$

Here, K_m is the magnitude scaling factor and K_f is the frequency scaling factor.

Find the new values of the circuit.

The new value of resistance is,

$$\begin{aligned} R' &= K_m R \\ &= (200)(2) \\ &= 400\ \Omega \end{aligned}$$

The new value of inductance is,

$$\begin{aligned} L' &= \frac{K_m}{K_f} L \\ &= \left(\frac{200}{10^4}\right)(1) \\ &= 0.02\ \text{H} \end{aligned}$$

The new value of capacitance is,

$$\begin{aligned} C' &= \frac{1}{K_m K_f} C \\ &= \frac{0.5}{(200)(10^4)} \\ &= 0.25\ \mu\text{F} \end{aligned}$$

Step 2 of 8

The new circuit after scaling is,

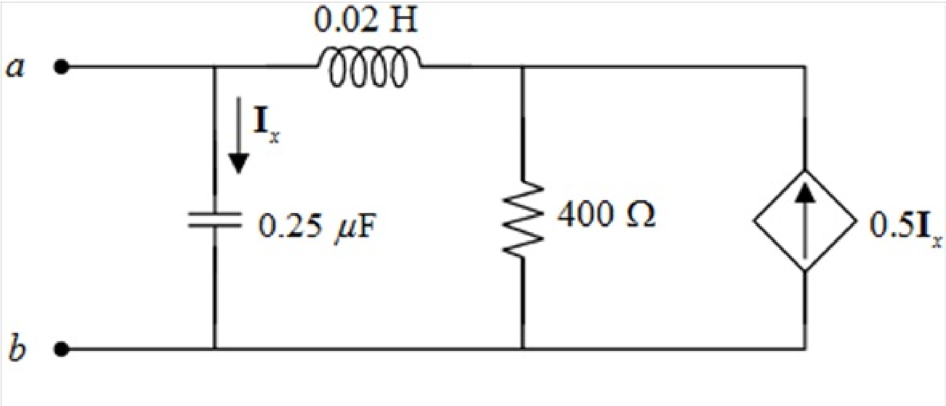


Figure 1

Step 3 of 8

(b)

Consider that the circuit operates at the angular frequency $\omega = 10^4 \text{ rad/s}$.

Calculate the values of circuit elements in frequency domain.

$$\begin{aligned} R' &= 400\ \Omega \\ X'_L &= j\omega L' \\ &= j(10^4)(0.02) \\ &= j200\ \Omega \end{aligned}$$

$$\begin{aligned} X'_C &= \frac{1}{j\omega C'} \\ &= \frac{1}{j(10^4)(0.25\times10^{-6})} \\ &= \frac{1}{j0.0025} \\ &= -j400\ \Omega \end{aligned}$$

Step 4 of 8

Apply a 1 A current source between the terminals of a and b and convert the circuit in to frequency domain.

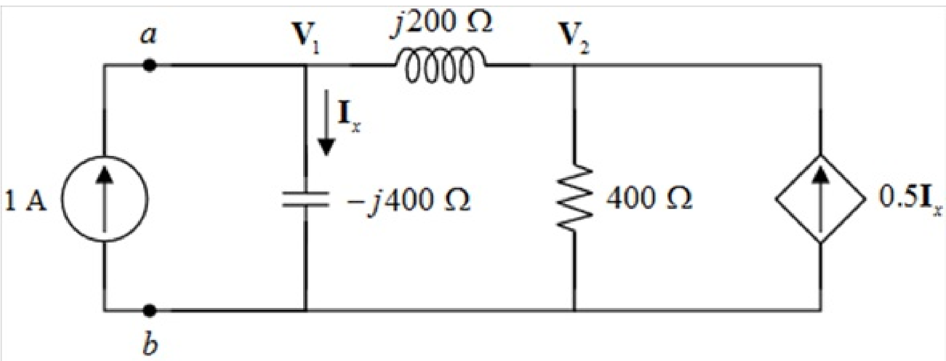


Figure 2

Step 5 of 8

Apply Kirchhoff's current law both the nodes in the circuit.

The node equation at $\mathbf{V}_1$ is,

$$1\text{ A} = \frac{\mathbf{V}_1}{-j400} + \frac{\mathbf{V}_1 - \mathbf{V}_2}{j200}$$

Simplify the equation.

$$1 = \mathbf{V}_1\left(\frac{1}{-j400} + \frac{1}{j200}\right) - \mathbf{V}_2 \frac{1}{j200}$$
$$\mathbf{V}_1\left(j0.0025 - j0.005\right) - \mathbf{V}_2\left(-j0.005\right) = 1$$
$$\mathbf{V}_1\left(-j0.0025\right) - 1 = \mathbf{V}_2\left(-j0.005\right)$$
$$\mathbf{V}_2 = \frac{\mathbf{V}_1\left(-j0.0025\right) - 1}{-j0.005}$$
$$= \frac{\mathbf{V}_1}{2} - j200$$

Step 6 of 8

The node equation at $\mathbf{V}_2$ is,

$$\frac{\mathbf{V}_2 - \mathbf{V}_1}{j200} + \frac{\mathbf{V}_2}{400} = 0.5\mathbf{I}_x$$

Here, the current $\mathbf{I}_x$ in the circuit is,

$$\mathbf{I}_x = \frac{\mathbf{V}_1}{-j400}$$

Therefore,

$$\frac{\mathbf{V}_2 - \mathbf{V}_1}{j200} + \frac{\mathbf{V}_2}{400} = 0.5\left(\frac{\mathbf{V}_1}{-j400}\right)$$

Simplify the equation.

$$\mathbf{V}_2\left(\frac{1}{j200} + \frac{1}{400}\right) - \frac{\mathbf{V}_1}{j200} = \mathbf{V}_1\left(\frac{0.5}{-j400}\right)$$
$$\mathbf{V}_2\left(\frac{1}{j200} + \frac{1}{400}\right) = \mathbf{V}_1\left(\frac{0.5}{-j400} + \frac{1}{j200}\right)$$
$$\mathbf{V}_2\left(-j0.005 + 0.0025\right) = \mathbf{V}_1\left(\frac{1.5}{j400}\right)$$

Step 7 of 8

Simplify the equation further to get the expression of $\mathbf{V}_1$.

$$\begin{aligned}\mathbf{V}_1 &= \frac{\mathbf{V}_2(-j0.005+0.0025)(j400)}{1.5}\\ &= \left(\frac{2+j1}{1.5}\right)\mathbf{V}_2\\ &= (1.34+j0.67)\mathbf{V}_2\end{aligned}$$

Substitute the expression of $\mathbf{V}_2$.

$$\begin{aligned}\mathbf{V}_1 &= (1.34+j0.67)\left(\frac{\mathbf{V}_1}{2}-j200\right)\\ &= (0.67+j0.34)\mathbf{V}_1-(-133.34+j266.67)\end{aligned}$$

Simplify the equation further to get the value of $\mathbf{V}_1$.

$$\mathbf{V}_1-(0.67+j0.34)\mathbf{V}_1 = -(-133.34+j266.67)$$
$$(0.34-j0.34)\mathbf{V}_1 = 133.34-j266.67$$

$$\begin{aligned}\mathbf{V}_1 &= \frac{133.34-j266.67}{0.34-j0.34}\\ &\approx 600-j200\\ &= 632.45\angle -18.435^\circ \text{ V}\end{aligned}$$

Step 8 of 8

Calculate the Thevenin's equivalent impedance of the circuit seen through the terminals a and b.

$$\begin{aligned}\mathbf{Z}_{\text{Th}} &= \frac{\mathbf{V}_1}{\mathbf{I}}\\ &= \frac{632.45\angle -18.435^\circ}{1}\\ &= 632.45\angle -18.435^\circ\ \Omega\end{aligned}$$

Thus, the Thevenin's equivalent impedance of the circuit seen through the terminals a and b is $\boxed{632.45\angle -18.435^\circ\ \Omega}$.